Title

"Comprehensive Hallucination Mitigation in Language Models Using Circuit Tracing via Transcoders/Crosscoders"

Background

Mechanistic Interpretability (MI) [1], [2], [3] is a subfield of Explainable AI that aims at analysing neural network activations at the neuron level to identify explainable *features* in neural network activations. Transcoders [4], replaced *Sparse Autoencoders* (SAEs) as the state of the art for MI [5]. Anthropic introduced the concept of Circuit Tracing with *Cross-Layer Transcoders* (CLTs) [6] with a companion paper [7] applying their circuit tracing techniques. Circuits refers to computational subgraphs responsible for specific model behaviours, such as implementing algorithms, encoded in the weights of machine learning models [3], [4], [6]. In their most recent research, Gurnee et al. [8] applied the method, but using *Weakly Causal Crosscoders* (WCCs), another promising MI neural network architecture.

Hallucinations in a language model refer to instances when the model responds to a prompt with made-up claims or concepts, without regard for context or factual accuracy [9], [10]. This prevents application of LLMs in any information critical domain. Current methods to mitigate this include *Retrieval Augmented Generation* (RAG) [11] and Reasoning Models [12], which cause the model to be grounded to some truth and reason with their response to ensure accuracy. Unfortunately these methods have not been able to completely solve the issue [13], which is where MI offers a promising means to comprehensively understand and subsequently prevent hallucinations more reliably [14], [15].

Research Gap Being Addressed

Hallucination remains a persistent issue in LLMs. While Lindsey et al. [7] briefly evaluates CLT circuit tracing on hallucinations, most other promising MI research [14], [15], [16], [17], [18] focuses on SAEs or prior methods CLTs/WCCs improve on. Further, this research proposes means to address unfaithfulness in CLTs and verify it's functioning experimentally. Finally we aim to produce a means to mitigate reliably by steering the model to not hallucinate.

Proposed Idea

1. Apply the circuit tracing methodology from Ameisen et al. [6] to an LLM for a set of prompts aimed at inducing hallucination (with or without reasoning and RAG) to analyse them for pathways indicative of hallucination.
2. Further using these circuits *steering* the LLMs away from hallucination pathways to increase reliability and confidence in their generated outputs.
3. From Lange et al. [19] CLTs have a problem wherein they learn results from one layer to another without considering the mechanistic contributions of intermediate layers. This poses a problem for hallucination detection specifically, to mitigate this I propose:
   (a) Using Per-Layer Transcoders and Weakly Causal Crosscoders as well as CLTs.
   (b) Limit cross layer connections in the CLT, modifying the architecture from Ameisen et al. [6].
   (c) Add an auxiliary faithfulness loss, as suggested by Lange et al. [19] but use Centred Kernel Alignment [20] instead of correlation.

Proposed Methodology

1. **Dataset Curation:** Review literature on prompting for hallucination - find factors that vary with hallucination; long-text generation, short-text generation, with/without reasoning/RAG etc. Use to develop comprehensive prompt dataset. Constrain task to one specific domain at this early stage, for ex. prompts corresponding to medicine. *Evalulation metric: Hallucination Induction Success Rate; how many 'hallucination' prompts consistently provoke hallucination in a given number of trials.*

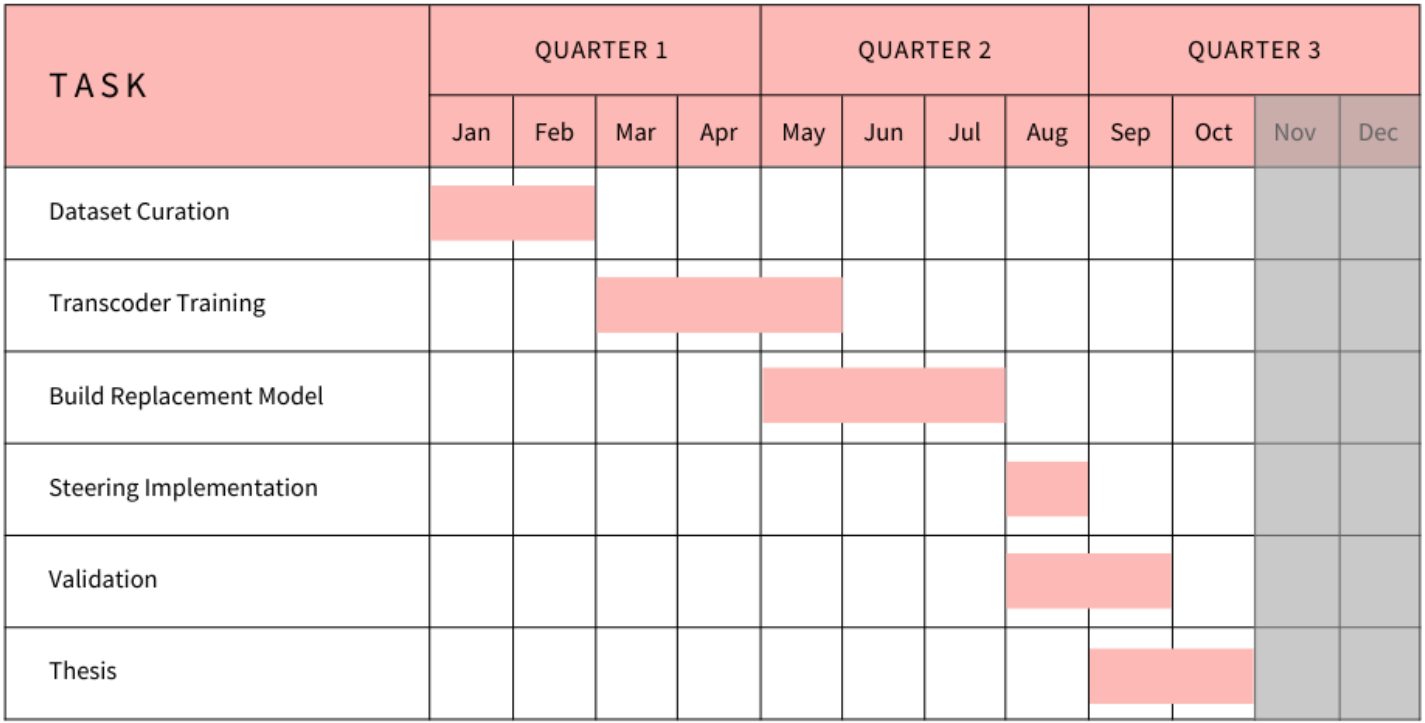
2. **Transcoder/Crosscoder Training:** They are trained to optimise reconstruction loss when replacing MLP (FFN) layers in the LLM. Further optimise for 'faithfulness' to intermediate layers. Construct replacement model of LLM from the trained CLT/WCC/PLT. *Evalulation metric: Reconstruction Loss between actual MLP output and Replacement MLP output (MSE + L1 (sparsity) penalty.)*

3. **Circuit Tracing:** Following Ameisen et al. [6] methodology to visualise attribution graphs, and identifying patterns specific to hallucination (or lack thereof). ***Using their publicised technology is what mainly permits meeting the 10 month time constraint.*** *Evaluation Metric: KL Divergence/Explained Variance Ratio to describe the faithfulness/explain-ability of replacement model output to original model. Logit Difference Ratio for confirming presence of hallucination features.*

4. **Steering:** Should we succeed above, we want to use the features identified and clamp them to steer models in a certain direction. This should force them to not hallucinate in a given domain of knowledge. *Evaluation Metric: Visual inspection of attribution graph. Circuit ablation to see effects on hallucination rate across trials from step one.*

Sample Research Statement Timeline

Gantt Chart- Assuming this starts in Jan of some year.



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